

Ground truth: benign for all four cases.

Std VIB assigns > 80% melanoma probability under any heavy degradation; QCTS (ours) cuts overconfidence by 12–14 pp without retraining.

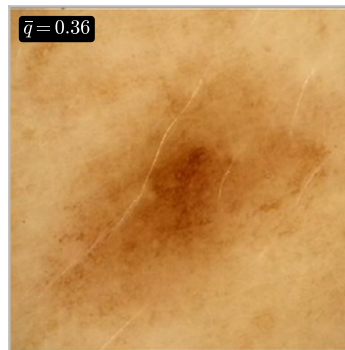
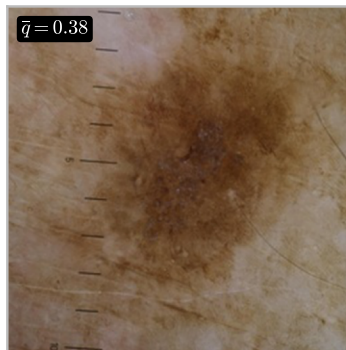
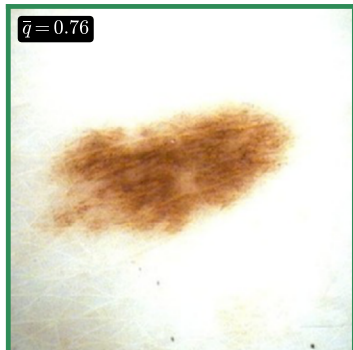
High Quality

Blur

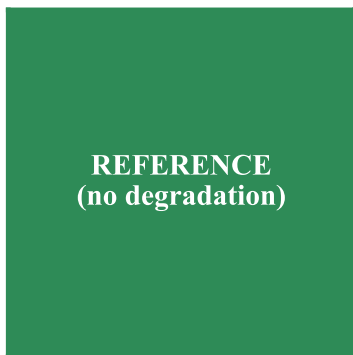
Colour Shift

Combined

Original input



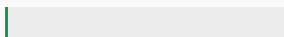
Heavy degradation



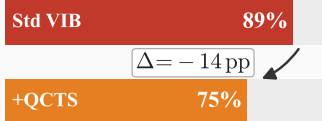
Confidence P(melanoma)

Ground truth: benign · P(mel) should be ≈ 0

Std VIB P(mel) = 1%



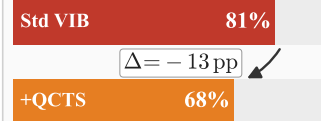
✓ Well-calibrated benign



× Overconfident on benign lesion



× Overconfident on benign lesion



× Overconfident on benign lesion